Econometrics II (JEB110)

Home Assignment 1

Deadline: October 29, 2025, 11:59 AM (NOON)

Dear students,

You are encouraged to work in groups of two people. In that case hand only one solution with names of both participants on the first page. In case you cannot find any colleague to form a group with, you can write an email to *oliver.skultety@fsv.cuni.cz* and I will match you with somebody. Submit your solution electronically via SIS. Pack all submitted files into a zipped folder named **"Surname1_Surname2_HA1"**.

The text itself can be written in any text-editing software of your choice (e.g., MS Word, LATEX). Attach both R "log" file with results and "rmd" file with the programming code to your assignment. If using another statistical software, attach the corresponding file(s). Use comments in your programming code to describe in few words what each command means. In case of problems with R, consult the official online manual and help module.

Keep in mind that not only the correctness of the results is assessed, but also the text editing quality is taken into consideration. Answer only what you are asked for.

Note that the use of **Artificial Intelligence (AI)** tools is only allowed to improve your writing style and grammar or to generate inspiration. However, treat the AI tools' output carefully and with caution, respecting the principles of academic integrity and the requirement to produce your own original work that is based on reliable and verifiable sources of information. Simple copying of the text generated by the AI tools is not permitted. Consult the full text of AI guidelines here.

Problem 1

In this problem, you are asked to compare the variance of OLS estimators coming from models that differ in the frequency of the used data. In each case, we explain the dependent variable y_t by the explanatory variable x_t. The variable y_t is observed for 31 time periods. However, the variable x_t, is observed only every 10th period: $x_1 = 2$, $x_{11} = 3$, $x_{21} = 4.5$ and $x_{31} = 5$.

In **Model 1**, we impute the missing values of the dependent variable by linear interpolation. This means that $x_2 = 2.1$, $x_3 = 2.2...$, $x_{12} = 3.15$,... and the estimated model is

$$y_t = \beta_0 + \beta_1 x_t + u_t, \tag{1}$$

with $t = \{1, 2, ..., 31\}$. u_t are i.i.d. errors with zero mean and positive standard deviation σ.

In **Model 2**, we reduce the frequency of the data to the frequency with which the explanatory variable is observed simply by using every 10th observation of the dependent variable.

$$y_s = \beta_0 + \beta_1 x_s + u_s, \tag{2}$$

with $s = \{1, 11, 21, 31\}$. u_s are i.i.d. errors with zero mean and positive standard deviation σ.

In **Model 3**, we reduce the frequency of the data by computing 10-period averages. This means that the following model is estimated:

$$Y_r = \beta_0 + \beta_1 X_r + U_r, \tag{3}$$

with $r = \{1, 2, 3\}$. The value of $Y_r = \sum_{t=1+10*(r-1)}^{1+10*r}(y_t)$, the value of $X_r = \sum_{t=1+10*(r-1)}^{10*r}(x_t)$, and $U_r = \sum_{1+10*(r-1)}^{10*r}(U_t)$. This is a common approach in macroeconomic analysis, where we might not observe monthly data, but only sums or averages over the whole year.

a) Calculate the variance of the OLS estimator as a function of the residual variance for each of the models.

b) Compare the variance of the OLS estimator of the first model with the second model.

c) Compare the variance of the OLS estimator of the first model with the third model.

d) Which of the three approaches would you choose?

Problem 2

Let us analyze which macroeconomic factors contribute the the observed increase in housing prices. Think about it as your contribution to the current political debate concerning the inaccessibility of housing. At your disposal is a quarterly dataset of the housing price index in the Czech Republic between 1Q 2015 and 4Q 2024. The dataset is augmented with several macroeconomic indicators for the same period. You are free to add more variables if you find them relevant.

The solution to this problem should be submitted in the form of a mini-project report (max 5 pages, preferably less, including graphs and tables) with an introduction, main text, and conclusion. **Do not provide the solution in numbered or bullet points**, but write a consistent text with equations, graphs, and tables. Do not include R-code in the text; submit it as an attachment.

You are provided with a dataset that contains quarterly information on housing prices, the number of constructions started, the nominal GDP, the inflation rate, and the population size. You are tasked with building an econometric model that can be used to explain why housing prices are growing. Use the following points as an indicative checklist, but write your solution as a continuous text, not a reaction to each point.

1. Think about the housing prices. Use your intuition and/or the economic theory to argue what might affect them.

2. Open the *Ecox2_HW1_data.csv* file. Get to know the data and make a primary analysis.

   2.1. What is the structure of the data (cross-sectional, time series, panel data)? If cross-sectional or panel, what is the unit of observation? If time series or panel, what is the frequency of observations?

2.2. Do you find all the relevant variables in the dataset? Is there a need to add something?

2.3. Prepare a summary table of the data. Are there any missing values? Are there any outliers or extreme values? What would you do about them?

2.4. Draw graphs of the most important variables and their evolution over time.

2.5. Is there any seasonality in the data? Are any variables trending? Do you expect any issues with stationarity?

3. It is often a good idea to start with a simple bivariate model. For example, check whether housing prices depend on GDP. Build a simple regression model and estimate it.

3.1. Can the model be estimated by OLS? Do you expect the OLS estimates to be unbiased? Consistent? Efficient? Why? What can be done to improve the analysis? Provide the necessary tests or theoretical arguments to support your statements.

3.2. Are any of the variables trending? Try to account for time trends in the analyzed variables. Compare the results to a regression without a trend.

3.3. Build a final regression model and estimate it. Are the standard errors of your model computed correctly? Use the relevant tests. Update the standard errors if necessary.

4. Once you experiment with a simple model, it is time to build the full regression model.

4.1. You may use the variables included in the data or any other variables that you find relevant. If you use additional variables, provide the sources.

4.2. Justify the use of the variables and their form (lags, logs, per capita adjustments, adjustment for inflation, etc.). Are the OLS assumptions satisfied? Provide the necessary tests.

4.3. You might estimate one specification (i.e., a model with one set of variables) or several alternative specifications (i.e., models with different sets and/or functional forms of the included explanatory variables). If estimating several alternative specifications, report them in one table with each column representing one approach. Compare the results and comment on potential differences.

5. Do not forget to add a discussion of your results. Do you trust them? Why? What could be improved if you had more time and/or access to better data? What is the policy implication from the presented results?